ODOO ERP FUNCTIONAL ANALYSIS CASE STUDY

Inventory & Procurement Process Improvement

Apex Components Pvt. Ltd. (Hypothetical)

Automotive Component Manufacturing

PROJECT DETAILS

ERP Platform	**Odoo Community Edition**
Modules	**Inventory and Purchase**
Project Type	**Functional Analysis Case Study**
Demonstrated Product	**Steel Sheets**

CONTENTS

1. Project Summary • 2. Business Requirements • 3. Process Analysis
4. Gap Analysis • 5. UAT Test Cases

1. Project Summary

Inventory & Procurement Process Improvement – Odoo ERP Functional Analysis

1.1 Executive Summary

This project evaluates how Odoo ERP can improve inventory visibility and procurement coordination for a hypothetical automotive component manufacturer.

The implementation demonstrates integrated Inventory and Purchase workflows using Odoo Community Edition.

1.2 Business Challenge

The assumed existing business process relied on manual stock tracking and reactive procurement, creating risks of inventory discrepancies and raw material shortages.

1.3 Implemented Solution

- Configured raw material product records.
- Created supplier records.
- Configured inventory reordering rules.
- Generated replenishment proposals.
- Created and confirmed purchase orders.
- Validated warehouse goods receipts.
- Verified inventory quantity updates.

1.4 Demonstrated Results

The Steel Sheets scenario demonstrated the following:

100	85	15	85	100
Initial stock	Consumed	Remaining	Replenishment	Final stock

Metric	Result
Initial stock	100 units
Minimum stock	20 units
Maximum stock	100 units
Simulated consumption	85 units
Remaining stock	15 units
Replenishment proposal	85 units
Final stock	100 units

1.5 Business Value

The configured workflow demonstrates improved stock visibility, structured replenishment, procurement traceability and inventory updates following goods receipt.

> **Note:** No quantified operational savings or real-world production improvements were measured.

1.6 Project Limitations

- Hypothetical company and business scenario.
- Sample product and supplier records.
- One raw material completed through the full replenishment cycle.
- No manufacturing or accounting implementation.
- No custom code or external system integrations.

1.7 Tools and Skills

Odoo Community Edition • Inventory Management • Purchase Management • Business Requirement Analysis • Process Mapping • Gap Analysis • Functional Configuration • UAT

1.8 Conclusion

The case study demonstrates how standard Odoo Inventory and Purchase functionality can support a structured raw material replenishment and procurement process.

2. Business Requirements Document

Project information, objectives, requirements and scope

2.1 Project Information

Item	Detail
Project	Inventory & Procurement Process Improvement
Client	Apex Components Pvt. Ltd. (Hypothetical)
Industry	Automotive Component Manufacturing
ERP	Odoo Community Edition
Modules	Inventory and Purchase
Project Type	Functional Analysis Case Study

2.2 Business Background

Apex Components Pvt. Ltd. manufactures automotive components using raw materials such as steel sheets, aluminium rods and copper wire.

The company currently relies on manual inventory records and reactive procurement processes, resulting in stock discrepancies, delayed purchasing and potential production interruptions.

2.3 Business Problem

- Inventory quantities are not consistently updated.
- Raw material shortages are identified late.
- Procurement requests are initiated reactively.
- Purchase order tracking lacks centralized visibility.
- Incoming material records are not consistently synchronized with inventory.

2.4 Business Objectives

- Improve raw material stock visibility.
- Establish minimum and maximum inventory thresholds.
- Identify replenishment requirements.
- Standardize vendor and purchasing workflows.
- Ensure inventory updates after goods receipt.

2.5 Business Requirements

ID	Requirement	Priority
BR-01	Maintain product and stock records	High
BR-02	Configure minimum and maximum stock levels	High
BR-03	Identify products requiring replenishment	High
BR-04	Generate purchase RFQs	High
BR-05	Confirm purchase orders	High
BR-06	Validate incoming goods receipts	High
BR-07	Update inventory after receipt validation	High

2.6 Project Scope

In Scope

- Inventory product configuration
- Vendor management
- Reordering rules
- RFQ generation
- Purchase order confirmation
- Goods receipt and stock verification

Out of Scope

- Manufacturing execution
- Accounting implementation
- Sales management
- Custom development
- External supplier integration

2.7 Expected Business Benefits

- Improved inventory visibility
- Structured procurement coordination
- Better stock replenishment control
- Improved purchase order traceability
- Reduced reliance on manual inventory tracking

2.8 Assumptions

> This is a hypothetical business case study. Operational problems and the existing manual workflow are assumed for analysis purposes. Odoo configuration and transaction results are demonstrated using a sample dataset.

3. Business Process Analysis

AS-IS workflow versus the proposed TO-BE Odoo workflow

3.1 AS-IS Process – Existing Workflow

1	Warehouse staff maintain raw material quantities manually.
2	Stock records are updated periodically.
3	Procurement identifies shortages after stock levels become insufficient.
4	Purchase requests are communicated manually.
5	Purchase orders are created and tracked separately.
6	Received materials are manually recorded in inventory records.

Identified Bottlenecks

- Delayed stock visibility
- Inventory discrepancies
- Reactive procurement
- Limited purchase order tracking
- Risk of production material shortages

3.2 TO-BE Process – Proposed Odoo Workflow

1	Maintain raw material records in Odoo Inventory.
2	Configure minimum and maximum stock levels.
3	Monitor stock availability.
4	Identify replenishment requirements.
5	Generate RFQs through Odoo Purchase.
6	Confirm purchase orders.
7	Receive materials through warehouse receipts.
8	Validate receipts to update inventory quantities.

3.3 Demonstrated Business Scenario

Product: Steel Sheets

Step	Quantity
Initial inventory	100 units
Minimum stock	20 units
Maximum stock	100 units
Simulated consumption	85 units
Remaining stock	15 units
Replenishment proposal	85 units
Confirmed purchase quantity	85 units
Final inventory	100 units

3.4 Process Outcome

The demonstrated workflow connects inventory replenishment with procurement and goods receipt, providing a structured method for managing raw material availability.

4. Gap Analysis

Existing process versus proposed Odoo solution

4.1 Existing Process vs Proposed Odoo Solution

Business Area	AS-IS Situation	Proposed TO-BE Solution
Inventory tracking	Manual records	Centralized Odoo stock records
Stock monitoring	Periodic checks	System-based quantity visibility
Replenishment	Reactive purchasing	Configured reordering rules
Procurement	Manual coordination	Integrated RFQ and PO workflow
Goods receipt	Manual updates	Validated warehouse receipts
Order tracking	Disconnected records	Centralized purchase records

4.2 Functional Gap Assessment

ID	Identified Gap	Odoo Functionality
GAP-01	Inconsistent stock records	Inventory Management
GAP-02	Late shortage identification	Reordering Rules
GAP-03	Unstructured purchasing	Purchase RFQs
GAP-04	Limited order tracking	Purchase Orders
GAP-05	Delayed inventory updates	Goods Receipt Validation

4.3 Conclusion

Standard Odoo Inventory and Purchase functionality addresses the identified process gaps through centralized stock management, replenishment proposals, purchasing workflows and receipt validation.

The case study demonstrates these improvements through a controlled sample transaction.

5. User Acceptance Testing (UAT)

Test environment, scenarios and results

5.1 Test Environment

Item	Detail
ERP	Odoo Community Edition
Modules	Inventory and Purchase
Test Product	Steel Sheets
Test Vendor	Metro Steel Suppliers

5.2 Test Scenarios

Test ID	Test Scenario	Expected Result	Actual
TC-01	Create raw material product	Product record created	**Pass**
TC-02	Create vendor	Vendor record created	**Pass**
TC-03	Receive initial purchase of 100 units	Stock becomes 100	**Pass**
TC-04	Configure reordering rule	Min 20, Max 100 saved	**Pass**
TC-05	Adjust stock from 100 to 15	Stock becomes 15	**Pass**
TC-06	Generate replenishment proposal	85 units proposed	**Pass**
TC-07	Confirm purchase order	RFQ converted to PO	**Pass**
TC-08	Validate goods receipt	Stock becomes 100	**Pass**

5.3 Detailed Test Scenario: Replenishment

Precondition

- Current stock: 15 units
- Minimum quantity: 20
- Maximum quantity: 100

Execution

1	Open Inventory replenishment.
2	Identify the Steel Sheets shortage.
3	Execute replenishment.
4	Verify the generated RFQ.
5	Confirm the purchase order.
6	Validate the incoming goods receipt.

Expected Result

- Replenishment quantity: 85 units.
- Purchase order successfully confirmed.
- Goods receipt successfully validated.
- Final stock: 100 units.

5.4 Test Conclusion

The demonstrated end-to-end replenishment scenario passed validation using the sample Odoo database.

Note: Results apply to the demonstrated Steel Sheets scenario. Other products were created but not independently tested through the full procurement cycle.